

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q20-1623

Eagle 138 kV  
December 22, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1623. The Interconnection Customer proposes a 77.1 MW solar photovoltaic facility interconnecting at the Eagle 138 kV substation in the Pennsylvania area. The request was received on December 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 579 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$201,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STEWART MOUNTAIN DAM to SHEPAUG 138 kV  
Contingency: SUGAR HILL to RITTER PARK 345 kV  
Loading: 100.2% (Rating: 339 MVA)  
DFAX: 5.1%  
Pre-project loading: 99.0%

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                               | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------|----------------------------|---------------|-----------------------|
| STEWART MOUNTAIN DAM to SHEPAUG 138 kV | Reconductor line           | \$662,000     | 30% (\$201,000)       |
|                                        | Total allocated to project |               | \$201,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.